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Project:

Topic: Operating Systems - Project 1

Course:

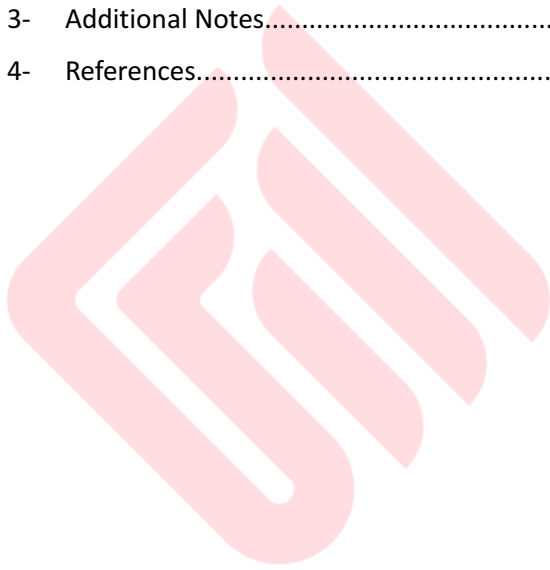
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1- Project Topic

Distributed Calculator System

This project aims to implement a distributed calculator system where four different mathematical operations (addition, subtraction, multiplication, and division) are handled by separate processes. The main program (calculator) communicates with these subprocesses, performs the operations, and displays the results to the user. The results are also saved to a file.

2- Tasks Completed During the Project

Main Program (calculator.c): The main program takes two numbers and an operation from the user, creates subprocesses for each operation, and sends the operation and operands to the appropriate subprocess. The results are received, displayed to the user, and passed to the saver program for storing in a file.

Mathematical Operations (addition.c, subtraction.c, multiplication.c, division.c): Each of these files implements a separate subprocess that performs one of the operations. They take two numbers as input, perform the operation, and send the result back to the main program.

Result Saving (saver.c): This program takes the calculated result as a parameter and appends it to a file (results.txt).

Communication and Process Management: Pipes were used for communication between the main program and subprocesses. Each operation has a dedicated pipe to ensure proper data flow.

Error Handling and User Feedback: Error handling was added for invalid inputs and division by zero, with appropriate error messages provided to the user.

3- Additional Notes

Distributed System Structure: In this project, each operation runs as an independent subprocess. The main program creates subprocesses for each operation and communicates with them via pipes. This ensures process isolation and modularity.

Pipe Communication: Data is transferred between the main program and subprocesses using pipes, which ensures smooth communication between processes.

Storing Results: The calculated results are not only printed to the screen but also saved to a file. This allows for persistent storage of all results.

Missing Features: The project could be improved by adding more robust input validation and error handling. Additionally, more flexibility could be added for switching between operations.

After accessing the main directory of the file on the console;

To compile, enter the "**make**" command.

After compiling, enter the command "**./calculator**" to run the program.

4- References

C Programming Language Documentation: General information and functions of the C programming language. Fundamental resources and documentation about C programming .

POSIX pipe() & fork() System Calls: Documentation on the pipe and fork system calls in the POSIX standard for inter-process communication and process management .

Linux System Programming: Detailed information on system programming and process management on Linux systems .

Lab Documentation: The project was also based on the provided lab documentation, which offered guidelines and explanations on how to structure the distributed calculator system and implement inter-process communication using pipes.

Video Tutorial: A useful video tutorial that provided insights into the project development process: [Unix Process in C](#).